Flash Attention

Standard attention computes the full $N \times N$ score matrix $S = QK^T$, writes it to HBM, reads it back for softmax, writes P back, reads it again for $O = PV$. This is memory bandwidth bound - most of the time is spent moving the data in / out of HBM, not doing the actual math.

Flash Attention avoids ever materializing the full $N \times N$ matrix in HBM. It tiles Q into row blocks and K/V into column-blocks, loads only small tiles into SRAM at a time, and computes the output incrementally using online softmax - maintaining a running max (m)

- running sum (l)
- running output accumulator (O) that

gets rescaled and merged as each new K/V block is processed. By the time all column-blocks have been seen, O/I is the exact, correct softmax $(QK^T)V$ - Computed without ever storing the full S or P matrix in HBM

Result: Same output as standard attention, but less HBM reads/writes → faster and memory usage scales linearly ($O(N)$) instead of quadratically ($O(N)^2$) in sequence length.

Flash Attention Backward.

lets do forward pass first then backward

$$Q = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} \qquad K = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix} \qquad V = \begin{bmatrix} 3 & 7 \\ 5 & 2 \\ 8 & 4 \end{bmatrix}$$

$S = QK^T$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} (2+2) & (4+1) & (2+1) \\ (1+4) & (2+2) & (1+2) \\ (3+2) & (6+1) & (3+1) \end{bmatrix}$$

$$S = \begin{bmatrix} 4 & 5 & 3 \\ 5 & 4 & 3 \\ 5 & 7 & 4 \end{bmatrix}$$

Now, $P = \text{Softmax}(S)$

steps : (1) find m, row max

$$m = \max(s_1, s_2, s_3)$$

(2) $\tilde{s}_i = s_i - m$

(3) $e_i = \exp(\tilde{s}_i)$

(4) $l = e_1 + e_2 + e_3$

(5) $p_i = \frac{e_i}{l}$ normalize.

$$S = \begin{array}{cl} \begin{bmatrix} 4 & 5 & 3 \\ 5 & 4 & 3 \\ 5 & 7 & 4 \end{bmatrix} & \begin{array}{l} - \text{row1} \\ - \text{row2} \\ - \text{row3} \end{array} \end{array}$$

row1

(1) $m = 5$

(2) $\hat{s}_i = [-1 \quad 0 \quad -2]$

(3) $e_i = [0.367, \ 1, \ 0.135]$

(4) $l = 1.5023$

(5) $p_i = \dfrac{e_i}{l} = [0.2442, \ 0.6656, \ 0.089]$

row2

$m = 5$

$\hat{s}_i = [0, -1, -2]$

$e_i = [1, \ 0.367, \ 0.135]$

$l = 1.5023$

$p_i = [0.6656, \ 0.2442, \ 0.089]$

row3

$m = 7$

$\hat{s} = [-2, \ 0, \ -3]$

$e_i = [0.135, \ 1, \ 0.0497]$

$l_i = 1.184$

$p_i = [0.1139, \ 0.8440, \ 0.0419]$

$$\therefore \quad P = \begin{bmatrix} 0.2442 & 0.6656 & 0.089 \\ 0.6656 & 0.2442 & 0.089 \\ 0.1139 & 0.8440 & 0.0419 \end{bmatrix}$$

$$O = PV \quad \underbrace{\begin{bmatrix} 0.2442 & 0.6656 & 0.089 \\ 0.6656 & 0.2442 & 0.089 \\ 0.1139 & 0.8440 & 0.0419 \end{bmatrix}}_{3\times3} \quad V = \underbrace{\begin{bmatrix} 3 & 7 \\ 5 & 2 \\ 8 & 4 \end{bmatrix}}_{3\times2}$$

$$\begin{bmatrix} 4.7807 & 3.4037 \\ 3.9396 & 5.5063 \\ 4.8970 & 2.6547 \end{bmatrix}$$

Finally now we go for backward pass.

The general pattern

$$dV = P^T . dO$$

$$dP = dO . V^T$$

ex.
$C = A . B$
$dB = A^T . dC$

Using jacobian

diagonal $(i = j)$: $P_i(1 - P_i)$

non-diagonal $(i \neq j)$ $-P_i . P_j$

$$\therefore \; J = \begin{bmatrix} P_i(1-P_i) & -P_i \cdot P_j & -P_i \cdot P_j \\ -P_i \cdot P_j & P_i(1-P_i) & -P_i \cdot P_j \\ -P_i \cdot P_j & -P_i \cdot P_j & P_i(1-P_i) \end{bmatrix}$$

$$\downarrow$$

$$J = \begin{bmatrix} P_1(1-P_1) & -P_1 \cdot P_2 & -P_1 \cdot P_3 \\ -P_2 \cdot P_1 & P_2(1-P_2) & -P_2 \cdot P_3 \\ -P_3 \cdot P_1 & -P_3 \cdot P_2 & P_3(1-P_3) \end{bmatrix}$$

row1 jacobian

$$P_1 = [0.2442 \quad 0.6656 \quad 0.089]$$

$$= \begin{bmatrix} 0.2442(1-0.2442) & -0.2442(0.6656) & -0.2442(0.089) \\ -0.6656(0.2442) & 0.6656(1-0.6656) & -0.6656(0.089) \\ -0.089(0.2442) & -0.089(0.6656) & 0.089(1-0.089) \end{bmatrix}$$

$$J_1 = \begin{bmatrix} 0.1846 & -0.1626 & -0.0217 \\ -0.1626 & 0.2226 & -0.0592 \\ -0.0217 & -0.0592 & -0.0810 \end{bmatrix}$$

$$ds_1 = J_i \cdot dp_1$$

$$ds_1 = [0.6638, \; -0.8540, \; 0.1971]$$

but Now thers a shortcut

$$ds_i = P_i(dp_i - c) \quad \text{where} \quad c = dp_i \cdot P_i$$

example for row 2.

$$P_2 = [0.6656, \quad 0.2442, \quad 0.089]$$
$$d_{P2} = [5.7, \ 6.6, \ 10.8]$$

① $C_2 = d_{P_2} \cdot P_2$

$$[5.7(0.6656) + 6.6(0.2442) + 10.8(0.089)]$$

$$C_2 = 3.7939 + 1.61172 + 0.9612$$
$$C_2 = 6.36682$$

② $d_{S_2} = P_2 \ (d_{P2} - C_2)$

$$= P_2 \ ([5.7 - 6.36, \ 6.6 - 6.36, \ 10.8 - 6.36])$$

$$= P_2 \cdot [-0.66, \ 0.24 \ \ 4.44]$$

$$= [0.6656, \ 0.2442, \ 0.089][-0.66, \ 0.24, \ 4.44]$$

$$d_{S_2} = [0.6656(-0.66), \ 0.2442(0.24), \ 0.089(4.44)]$$

$$= [-0.439, \ 0.0570, \ 0.39516]$$

Now, we do same for d_{S3}

And we finally get d_s matrix

$$dS = \begin{bmatrix} 0.6638 & -0.8540 & 0.1971 \\ -0.4438 & 0.0570 & 0.3947 \\ 0.1242 & -0.2610 & 0.1379 \end{bmatrix}$$

Final step dQ and dK.

do same as dS for dQ and dK.

we get,

$$dQ = \begin{bmatrix} -0.4813 & 0.6237 \\ 0.0649 & -0.3364 \\ -0.2599 & 0.0253 \end{bmatrix}$$

$$dK = \begin{bmatrix} 1.4104 & -0.1596 \\ -2.4680 & -0.9670 \\ 1.1897 & 1.3244 \end{bmatrix}$$